

Privacy Aware Computing (Project 1)

Tanvir Hossain (002728826)

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Seed Based Graph Deanonymization

Methodology

We develop a function for Seed-Based Graph Deanonymization f_{SBGD} where the input is two graphs: $G1, G2$ and a list of seed node pairs S_p .

In our approach, for a node $n \in G1$, we measure its neighbors who are included in the pairs of seed nodes $N(n_{s_{G1}}) = N(n) \cap \{n_{s1} | (n_{s1}, n_{s2}) \in S_p, n_{s1} \in G1\}$. Then we observe the corresponding seed nodes of $N(n_{s_{G1}})$, that is, $n_{s2} = \{n_{s2} | (n_{s1}, n_{s2}) \in S_p, n_{s2} \in G2\}$ and find all their neighbors $N(n_{s_{G2}}) = \bigcup N(u), \forall u \in n_{s2}$. Then we calculate the score function for $\forall m \in N(n_{s_{G2}})$ and get the set of scores $S_c(n)$.

$$score(n, m) = \frac{|N(n) \cap N(m)|_{(n,m) \in S_p}}{\sqrt{deg(n \in G1)} \times \sqrt{deg(m \in G2)}}$$

Then we find S_{max}, S_{max2} and the standard deviation σ from $S_c(n)$ and measure the eccentricity score.

$$ECCE = \frac{S_{max} - S_{max2}}{\sigma}$$

Given the eccentricity threshold η , if $ECCE \geq \eta$, then we consider m as the correspondence of n for which $score(n, m) = S_{max}$. Hence, the new seed pair (n, m) is included in seed node pairs, $S_p = S_p \cup \{(n, m)\}$. Note that, during the exploration of another node $(p \neq n) \in G1$ if we find the $score(p, m) > score(n, m)$, then we remove (n, m) from S_p and instead add (p, m) .

According to Algorithm 1, In the execution process, based on given seed pairs, we do our operations to get new seed pairs nS_p by calling the above procedure of Seed Based Graph Deanonymization f_{SBGD} . After the end of seed pair extraction, we measure the accuracy of our method.

Algorithm 1: Seed-Based Graph Deanonymization (Iterative Process)

Input: $G1, G2$, Initial seed pairs S_p

Output: Final seed pairs S_p

- 1 Initialize $nS_p = f_{SBGD}(G1, G2, S_p)$;
 - 2 Update $S_p = S_p \cup nS_p$;
 - 3 **while** $|nS_p| > 0$ **do**
 - 4 $nS_p = f_{SBGD}(G1, G2, S_p)$;
 - 5 $S_p = S_p \cup nS_p$;
 - 6 **return** S_p ;
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Table 1: Dataset Info

Graph 1	26956 (Nodes)	344379 (Edges)
Graph 2	27756 (Nodes)	348903 (Edges)
Number of Validation Seed Pairs	27770 (Given)	26948 (Valid)

Experiment and Results

The dataset information is given in Table 1, where the Valid Seed pair means 26948 seed pairs nodes exist in Two graphs. In our experiment, for a given validation seed pair, we use 20% of that as given seed pairs, and the rest of the 80% are used for final evaluation. We use the eccentricity

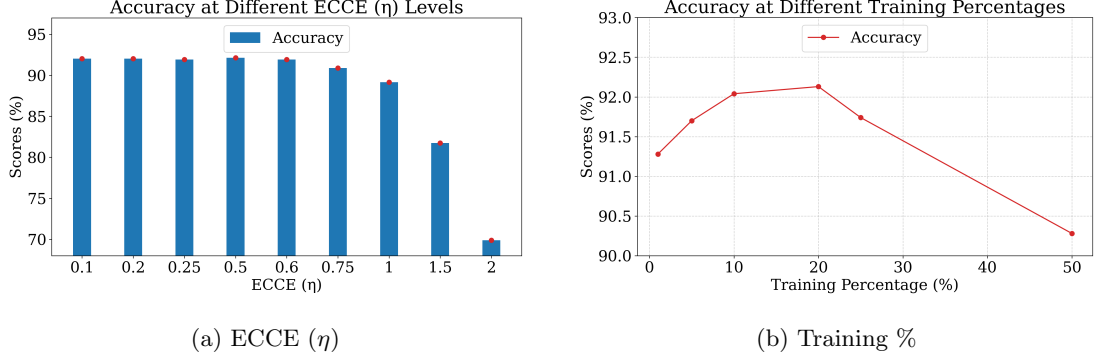


Figure 1: Accuracy for Change of Parameters

threshold $\eta = 0.5$ and get the accuracy score **92.13%**. For the change of η and training percentage %, we observe the performance of our method in Figure 1.

Seed Free Graph Deanonimization

Methodology

Node Embedding Based: In this method, we first tried to apply an unsupervised graph embedding method *Node2Vec* for getting the embedding of two graphs. Then we measure the cosine similarities between two graph nodes. for each node $n \in V_{G1}$, we measure to try to measure its embedding similarities (cosine) with each node in V_{G2} . We take the maximum similarity gained node $m \in G2$ as the seed pair of n . However, this procedure takes too much time and resources in the machine. With this approach, only one seed pair matches with validation seed pairs.

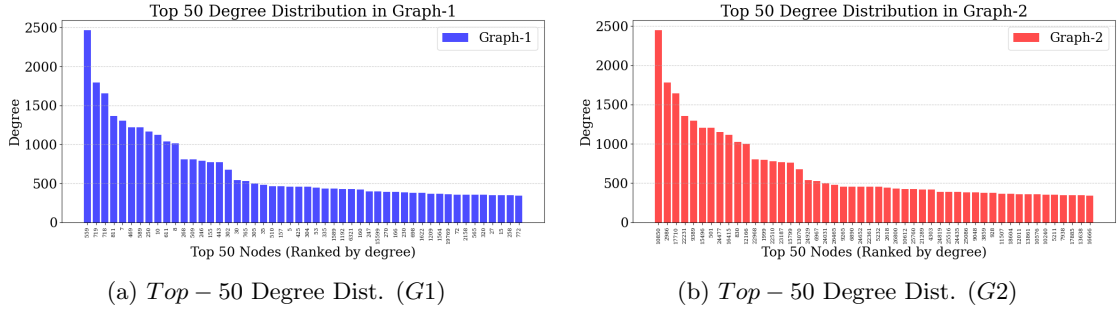


Figure 2: Degree Distributions of $G1$ and $G2$

Centrality Based: Next, we observe some properties of the two graphs such as degree distributions (in Figure 2), and node strength. Motivated by that observation, we use three structural graph property measuring functions to measure nodes' centrality scores: PageRank Centrality, Eigenvector Centrality, and Degree Centrality. After getting the centrality scores, we sort the nodes in descending order. After sorting, we align the nodes based on their rank in the sorted list. Finally, we measure the accuracy of our approaches with the validation set.

Cohesive Subgraph Based: In this approach we utilize two cohesive subgraph extraction algorithms: k -truss and k -core to measure the edge scores. K -truss focuses on the involvement of edges in triangles. If an edge is involved in $k - 2$ triangles then the edge trussness score is k . For example, in a triangle, each edge is involved in $(3 - 2 = 1)$ triangle, hence, the edge score is $k = 3$.

On the other hand, in a k -core subgraph, each node is involved in at least k number of incident edges. So for $k = 1$ to k_{max} , we extract the subgraphs and assign the k values to the subgraphs' edges. After getting the edge scores we measure each node's strength with the formula below. Then we rank the nodes based on the $Strength(u)$ score and pair up the similar index nodes as centrality-based measures.

$$Strength(u) = \sum_{v \in N(u)} weight(u, v)$$

Methods	# of Extracted Seed Pairs	Accuracy (%)
Node2Vec	3	0.01
PageRank Centrality	86	0.32
EigenVector Centrality	224	0.83
Degree Centrality	135	0.50
K-core	44	0.16
K-truss	29	0.10

Table 2: Result Table

Experiment and Results

The dataset is the same as the seed-based approach (Table 1). The performance of these approaches is not very promising because we do not have any prior information. In table 2, we show the evaluation scores in accuracy score for all the applied methods.

Integrate to the Seed Based Approach

As we measure the nodes' strength or centrality scores using the methods outlined in table 2, we observe several interesting properties during node pairing. In Figures 3(a) 3(b), and 3(c), during comparing the ranked nodes, the first 20 – 50 indexes nodes decently match. Even after matching 100 – 150 indexes we see nearly 40% nodes match accurately. So, if we use the first 25 – 150 of these matches as the seed pairs (S_p) to the Seed Based Graph Deanonimization method (f_{SBD}) then, **does that significantly improve the accuracy?**

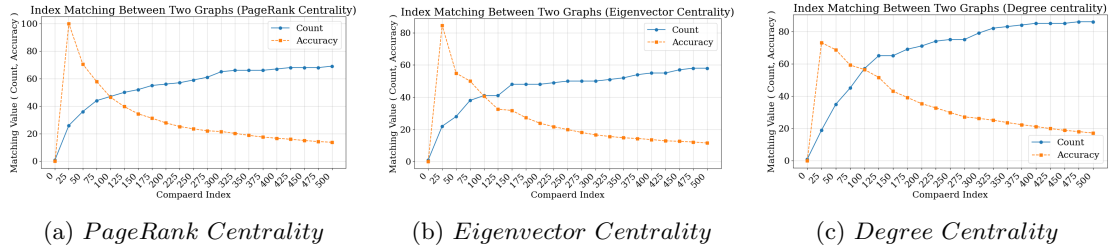


Figure 3: Measuring Accuracy and Matched Pair Count

The answer to the question is **yes**. It significantly improves the accuracy metric. In Figure 4, we demonstrate the integration result for different methods. We observe all the integrated methods (Fig 4(a)) achieve over 70% accuracy. Besides, in Figure 4(b), we show the changes of the result for given top indexes from the degree-centrality measure method. In the line plot, accuracy changes proportionally to increase the number of matched pairs. Our Integrated method achieves the best accuracy **88.88%** for *top* – 50 indices pairs from the PageRank measure.

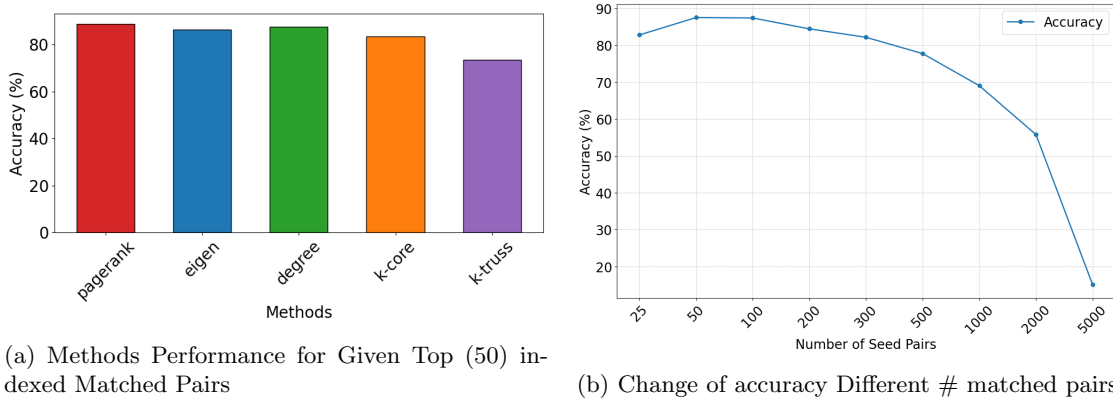


Figure 4: Performance After Utilizing Top-ranked Seed Pairs of Centrality-based Methods with Seed-Based Graph Deanonimization Technique